

University of Moratuwa
Faculty of Engineering
Department of Electronic & Telecommunication Engineering



EN1190 – Engineering Design Project

Team Supertronics

IoT-Based IV Fluid Monitoring System

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1. Problem Description

1.1. Problem

In many Sri Lankan hospitals, especially public and rural ones, the monitoring of IV fluid (saline) levels is still carried out manually by nurses and attendants. This manual method is time-consuming, error-prone, and highly inefficient in busy hospital environments. A neglected or unnoticed empty IV bag can lead to air embolism, delayed treatment, and patient discomfort, which may cause serious health complications and even fatalities in critical care units.

Hospitals in Sri Lanka, which are already facing workforce shortages and resource limitations, are further burdened by the constant need for staff to manually check saline levels for each bed. With over 100 beds per ward in some major hospitals, this becomes a logistical and operational bottleneck that impacts overall patient care quality.

Moreover, there is no centralized system currently in use that alerts medical staff in real-time when a saline bottle is empty or nearly empty. This results in poor prioritization and response time during emergencies.

1.2. Expected Goal

The primary goal of this project is to design and implement a low-cost, centralized IV fluid monitoring system that can be deployed in local hospitals to track the status of multiple IV bottles simultaneously.

This system should:

- Automatically detect when a saline bottle is near-empty or empty.
- Send real-time notifications to nurses or caregivers with the corresponding bed number.
- Be scalable (support 100+ beds), user-friendly, and cost-effective, allowing hospitals to improve patient safety and staff efficiency without a major financial burden.

1.3. Solution

The proposed solution is an IoT-based Centralized IV Fluid Monitoring System. The key components of the system include:

- A load cell sensor (with HX711 module) placed under each IV bottle to continuously measure the weight of the fluid.
- An ESP32 microcontroller to process sensor data and transmit it wirelessly to a central monitoring unit.
- A Wi-Fi or LoRa-based communication system (depending on hospital size and infrastructure).
- A mobile-optimized web dashboard that displays:
 - Real-time saline levels of all beds.
 - Visual and audio alerts when fluid levels reach critical thresholds.
 - Clear mapping between the saline bottle and patient bed number.

The system helps medical staff prioritize patient care efficiently and eliminates the risk of unnoticed saline depletion.

1.4. Justification of Selection

Manual saline level monitoring is outdated, unreliable, and labor-intensive, especially in emergency wards and ICUs. In contrast, our proposed system offers:

- Accurate, real-time measurements of fluid levels.
- Scalability for use in large hospital wards with minimal added cost.
- A centralized alert system that reduces workload and improves response time.
- Cost-effectiveness, using affordable components (load cells, ESP32) to deliver features on par with expensive foreign medical systems.

Local interviews with nursing staff and biomedical engineers confirmed that there is a high need and market potential for such a system. By digitizing a simple but critical task, our project directly contributes to better patient safety, operational efficiency, and the modernization of hospital infrastructure in Sri Lanka.

2. Product Feasibility

2.1. Technical Feasibility

We have successfully developed a fully functional prototype of our centralized IV fluid monitoring system. The core of the system is built around the ESP32-WROOM-32 microcontroller, which enables both sensor data acquisition and Wi-Fi communication. Each unit continuously reads the fluid weight using a 5kg load cell connected to an HX711 amplifier, ensuring high-resolution measurement of saline levels.

The ESP32 transmits this data in real-time to a Firebase-powered cloud dashboard, built using React, allowing nurses and medical staff to monitor all active IV devices from a single centralized interface. Each device is assigned a unique ID, and nurses can log which device is connected to which bed using the dashboard itself. This provides the flexibility to move devices between beds without any hardcoded assignments.

Power is supplied via a 230V to 5V converter, regulated down to 3.3V using an AMS1117 voltage regulator, providing stable operation for all components. A digital display is included on each device to show the saline level, offering redundancy and usability even without internet access.

We used SolidWorks to design a custom-fitted enclosure, and the PCB was designed using Altium Designer, ensuring precision, compactness, and ease of production.

2.2. Hardware Feasibility

The system is composed of readily available, cost-effective components that ensure reliability and scalability. Below are the key hardware components used in the device:

- 5kg Load Cell – for accurate measurement of the saline bottle's weight.
- HX711 Load Cell Amplifier Module – to amplify and digitize the load cell signal.
- ESP32-WROOM-32 Microcontroller – for data processing and Wi-Fi connectivity.

- Digital OLED Screen – to locally display real-time saline weight.
- AMS1117 3.3V Voltage Regulator – to ensure a stable operating voltage for the ESP32.
- 5V DC Power Supply (from 230V AC) – reliable continuous power from wall outlets.

The hardware design ensures modularity, enabling each unit to function independently and communicate wirelessly with the central dashboard. The choice of components balances affordability, durability, and performance—making the solution suitable for wide deployment in Sri Lankan hospitals.

2.3. Software Feasibility

<i>Component</i>	<i>Technology</i>	<i>Purpose</i>
ESP32 Firmware	C++ (Arduino IDE)	To read load cell data, manage Wi-Fi, and send real-time data to Firebase
Backend Database	Firebase Realtime Database	To store weight data and device info in real time
Frontend Dashboard	React.js (JavaScript)	To display real-time fluid levels and device status to hospital staff
Authentication	Firebase Authentication	To secure access and allow authorized users (nurses, admins)
Alert System	JavaScript & Firebase Triggers	To trigger sound and color-coded alerts based on threshold values

3. Product Applications

The IV fluid monitoring system is designed for continuous, real-time tracking of saline (IV) fluid levels in healthcare environments. Its purpose is not only to reduce manual workload but also to strengthen the reliability of patient care delivery through intelligent automation.

Primary Use Case

Healthcare workers place saline bottles on the load-cell-enabled device, which then begins to continuously measure and display the remaining fluid volume. The tare button is pressed prior to hanging a new bottle to reset the device’s weight reading. The on-device display shows live values for immediate local access, while the cloud-connected dashboard offers a centralized, ward-level view of all devices.

When a bottle is about to run dry or becomes empty, an automated alert is sent to the dashboard, enabling nurses and attendants to prioritize refills without constant manual checks.

Usage Environments

- Public and private hospital wards
- Intensive Care Units (ICUs)

- Emergency treatment rooms
- Home-based patient care
- Elderly care facilities and hospices
- Military and disaster relief medical units

Versatile Functionality

Bed Mobility: The device is not permanently assigned to a bed. Instead, each device has a unique ID, and the nurse logs which bed it's currently serving through the dashboard.

Offline and Online Readability: The built-in digital screen allows medical personnel to read values even in areas with poor internet.

Real-time Cloud View: All readings are synchronized via Wi-Fi to a Firebase-based dashboard, accessible from any device on the hospital's secure network.

Additional Scenarios

In busy hospitals during peak hours or pandemics, the system improves patient monitoring without requiring extra staff.

In training hospitals, this system can be used to teach and simulate proper IV handling and monitoring for medical interns.

During power outages, the display continues working as long as the main power supply is backed up (hospital UPS).

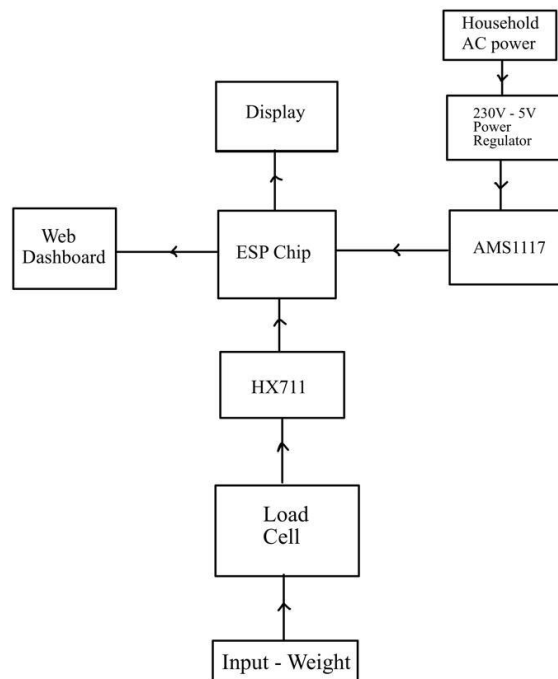
4. Product Specifications

- Type: IV Fluid Monitoring System
- Parameters Measured: Weight of saline bottle in grams (accuracy up to 1g)
- Display: Local digital screen
- Power Supply: 230V AC input, regulated to 5V DC
- Communication: Wi-Fi (via ESP32-WROOM-32)
- User Interface: Real-time dashboard via Firebase & React
- Device display at the bedside
- Onboard Tare (Zero) and Power On/Off buttons
- Material: Custom 3D printed enclosure (designed in SolidWorks)
- Dimensions: Height 15cm, Width 11cm, Weight 150g
- Scalability: Supports monitoring of 100+ beds
- ID System: Unique device ID displayed on dashboard; nurses log current bed assignment

5. Product Architecture

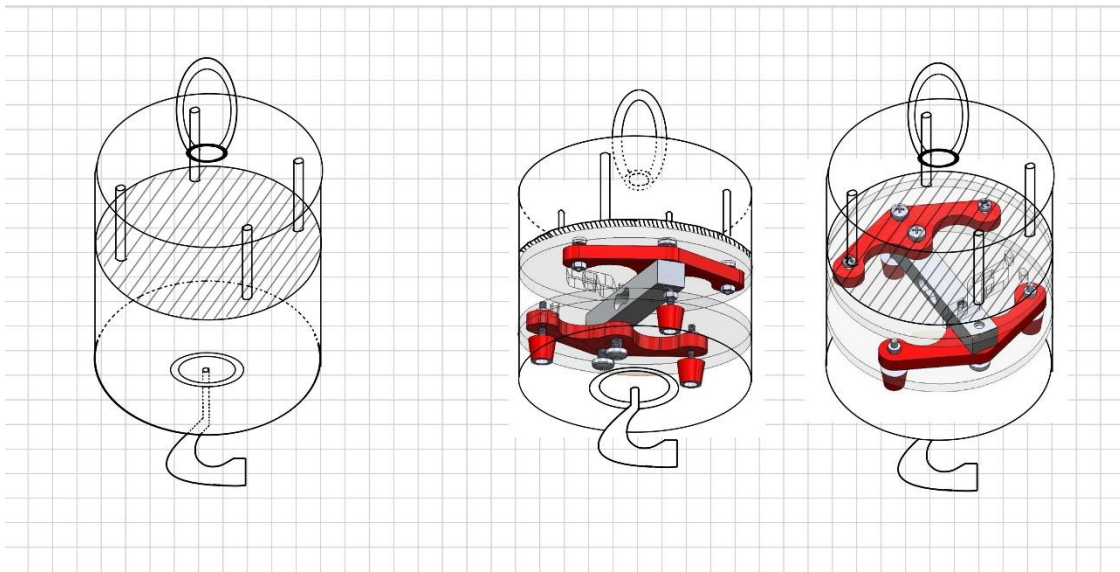
The architecture of the IV fluid monitoring system combines embedded hardware with cloud-based monitoring, designed for low cost and high usability. The key modules include:

- Power Supply: A 230V AC wall adapter provides 5V DC, which is stepped down to 3.3V using an AMS1117 voltage regulator to power the ESP32 microcontroller.
- ESP32-WROOM-32: Acts as the central processing unit, reading weight data from the sensor, handling Wi-Fi communication, and updating the Firebase database in real-time.
- Load Cell + HX711: A 5kg load cell measures the weight of the saline bottle with high precision, and the HX711 amplifier converts the analog signal to digital for the ESP32.
- Digital Display: A small screen (OLED) displays the current saline weight directly on the device.
- Tare Button: Used to reset the weight measurement to zero before a new saline bottle is added.
- Power Button: Allows users to safely switch the device on or off as needed.
- Enclosure: Custom-designed using SolidWorks and fabricated to securely house all components while allowing access to buttons and the display.
- Cloud Dashboard: Built with Firebase and React, it allows medical staff to monitor all devices, receive alerts, and log which device is in use at which bed.

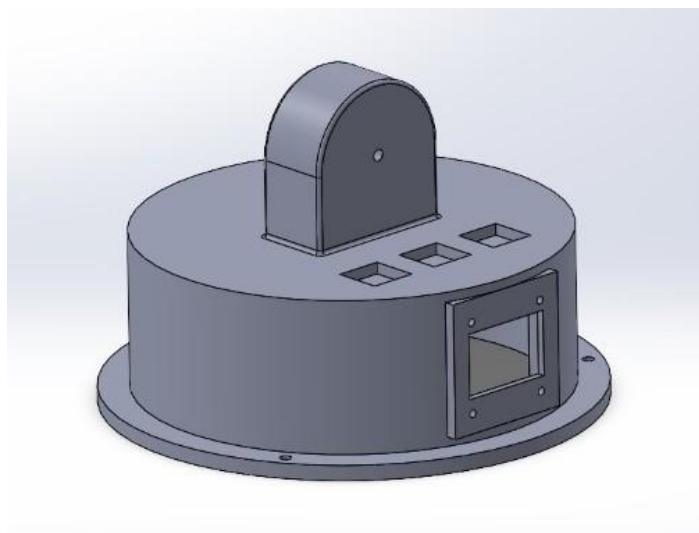


6. Sketches of the Product Enclosure

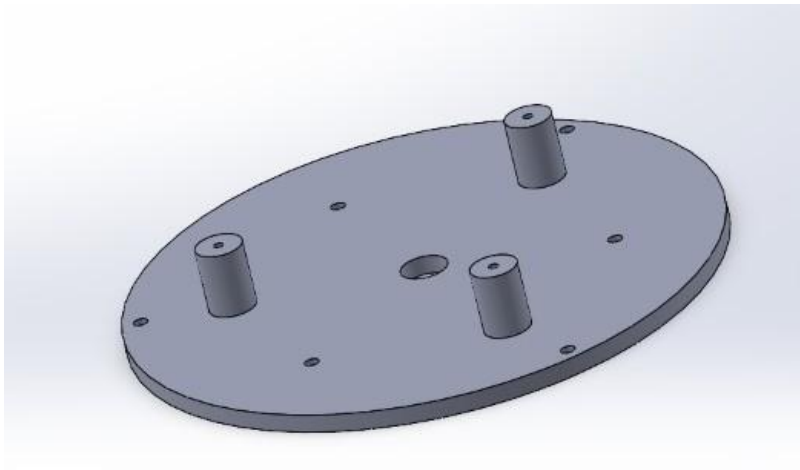
Initial Sketch



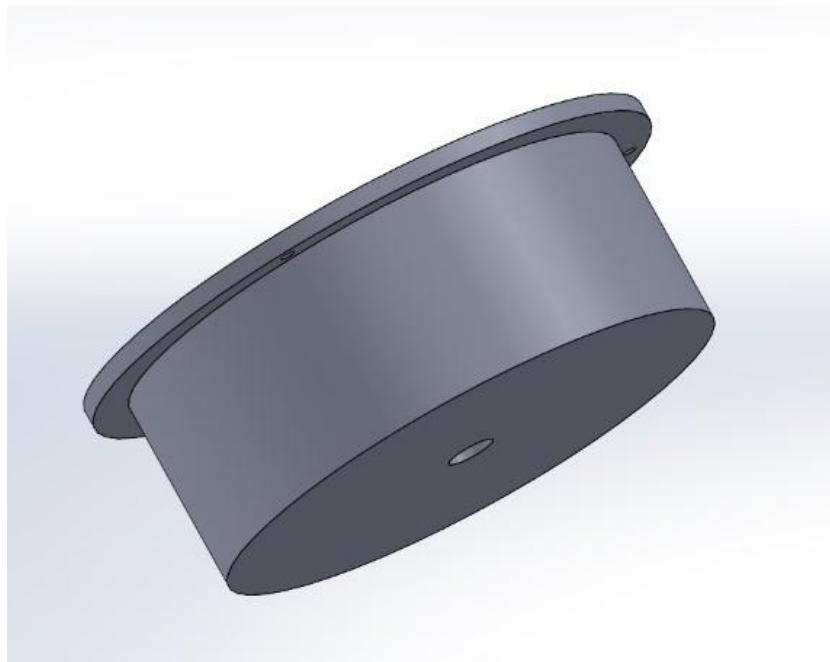
Top Part



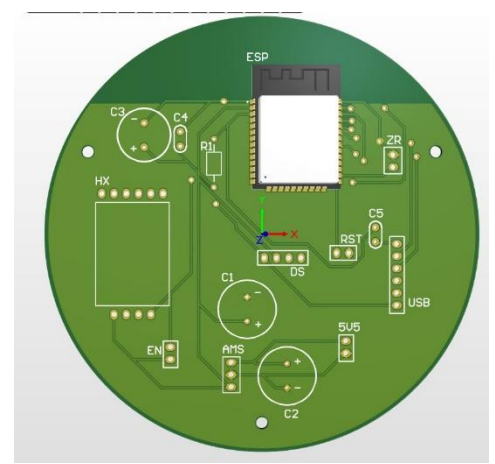
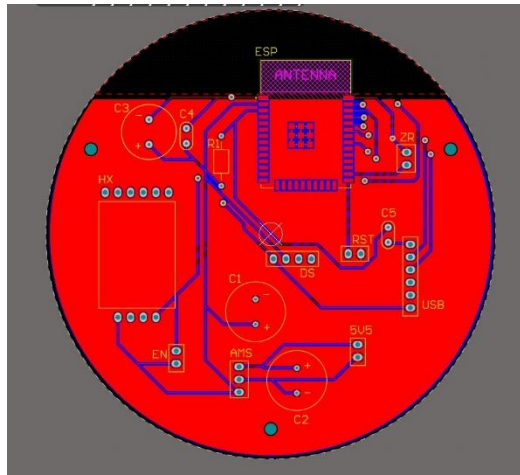
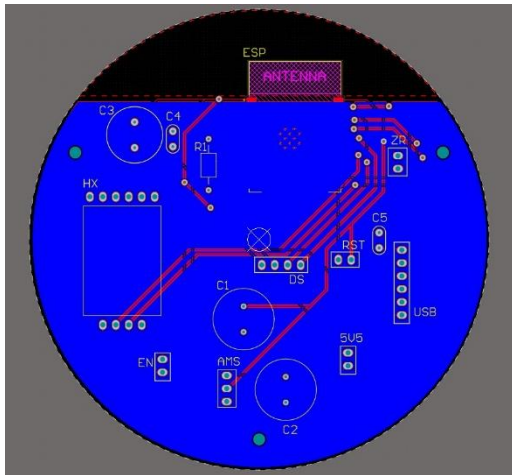
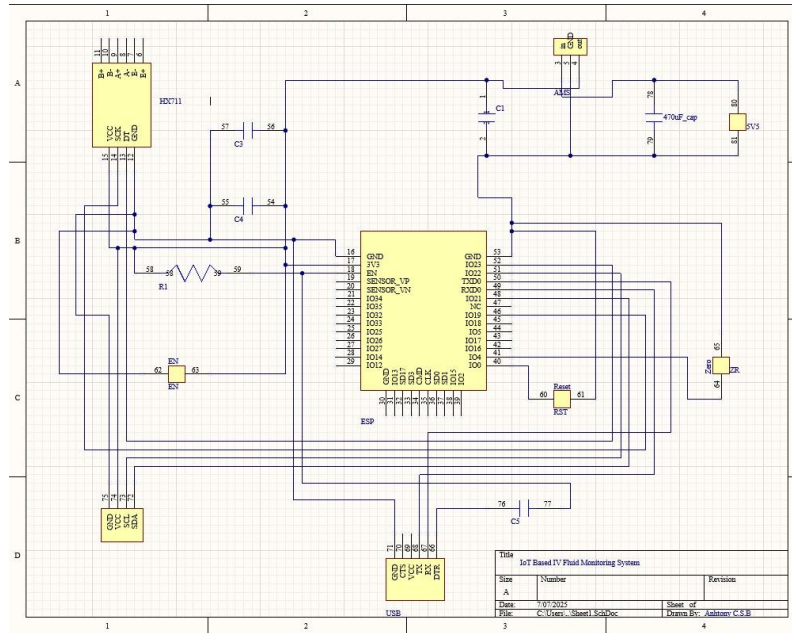
Middle Part



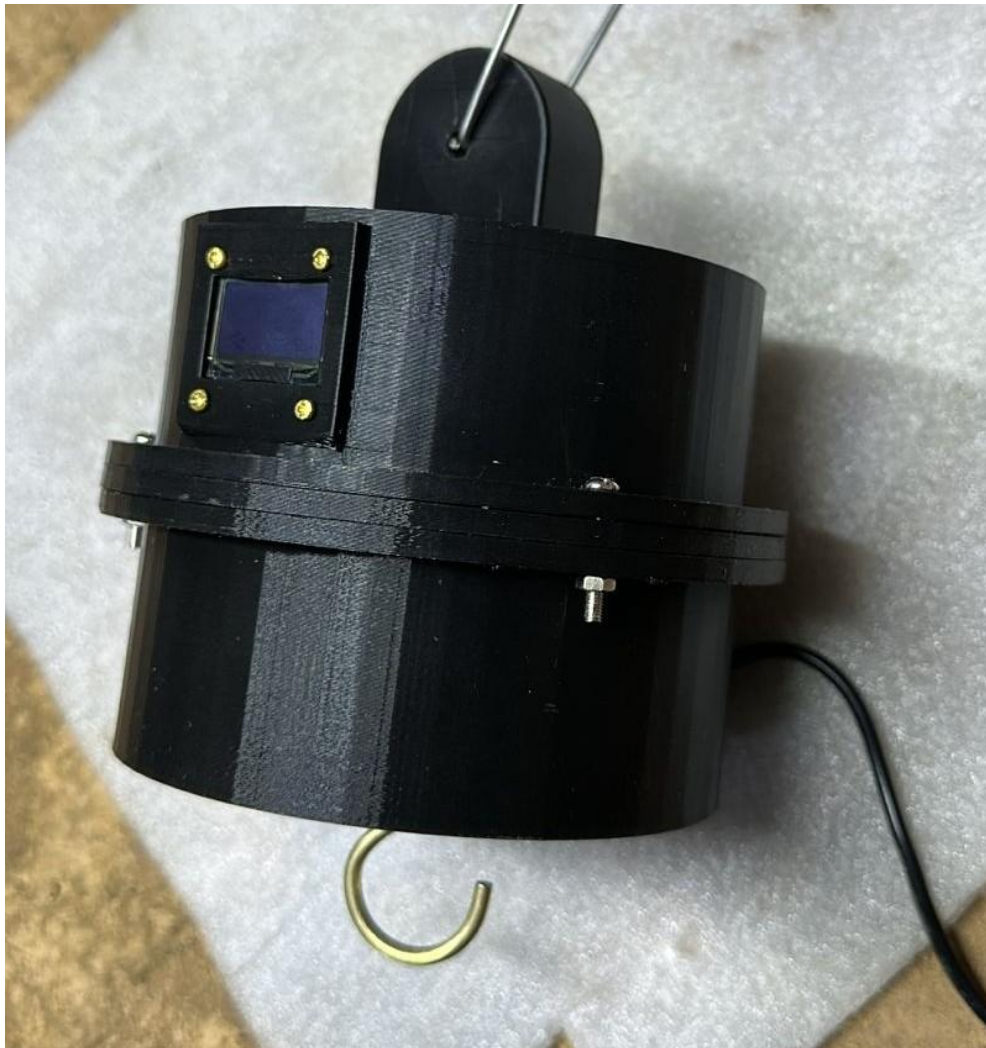
Bottom Part



7. PCB Design



8. Final Product



9. Market Analysis

There is a growing need in Sri Lanka's healthcare sector for automated, real-time monitoring systems to reduce staff workload and improve patient safety. Currently, IV fluid levels are monitored manually, which is time-consuming and error-prone, especially in overcrowded government hospitals.

Our system fills a critical gap by providing a low-cost, scalable solution for automated saline monitoring. Existing solutions in the global market are expensive (LKR 40,000–100,000 per unit), not tailored for local hospital settings, and largely unavailable in Sri Lanka.

Target Market

- Public and private hospitals
- Elderly care homes
- Rural clinics
- Home-based patient care
- Disaster relief and military hospitals

Business Models

- Direct B2B sales to hospitals
- Subscription model with maintenance support
- NGO/government-funded deployment in rural regions

Key Challenges

- Ensuring ease of use and hospital workflow integration
- Addressing data security in the dashboard
- Durability for long-term hospital use
- User training for staff

With high demand and no affordable alternatives in the local market, our system has strong potential to improve patient care and operational efficiency across Sri Lankan healthcare facilities.

10. Task Allocation

Name	Task
Anthony C.S.B.	PCB Design, Assembling, Circuit Analysis
Dharmakeerthi P.K.G.C.L.	Enclosure Design, Soldering, Assembling
Perera H.A.J.I.	Circuit Design, Load Cell Calibration, Component Selection
Wijethilaka J.S.	Web Interface & Alert System Design, Database Design, Assembling

11. Final Budget

Component	Cost (LKR)
PCB	400
ESP32	1400
Load Cell	350
Load Cell Bracket	1300
Enclosure	6100
HX711 Module	150
Display (OLED)	800
Other Components	500
Total	11000